

Physical Chemistry (II)

Lecture 21

Diatomic Molecules



Born-Oppenheimer Approx.

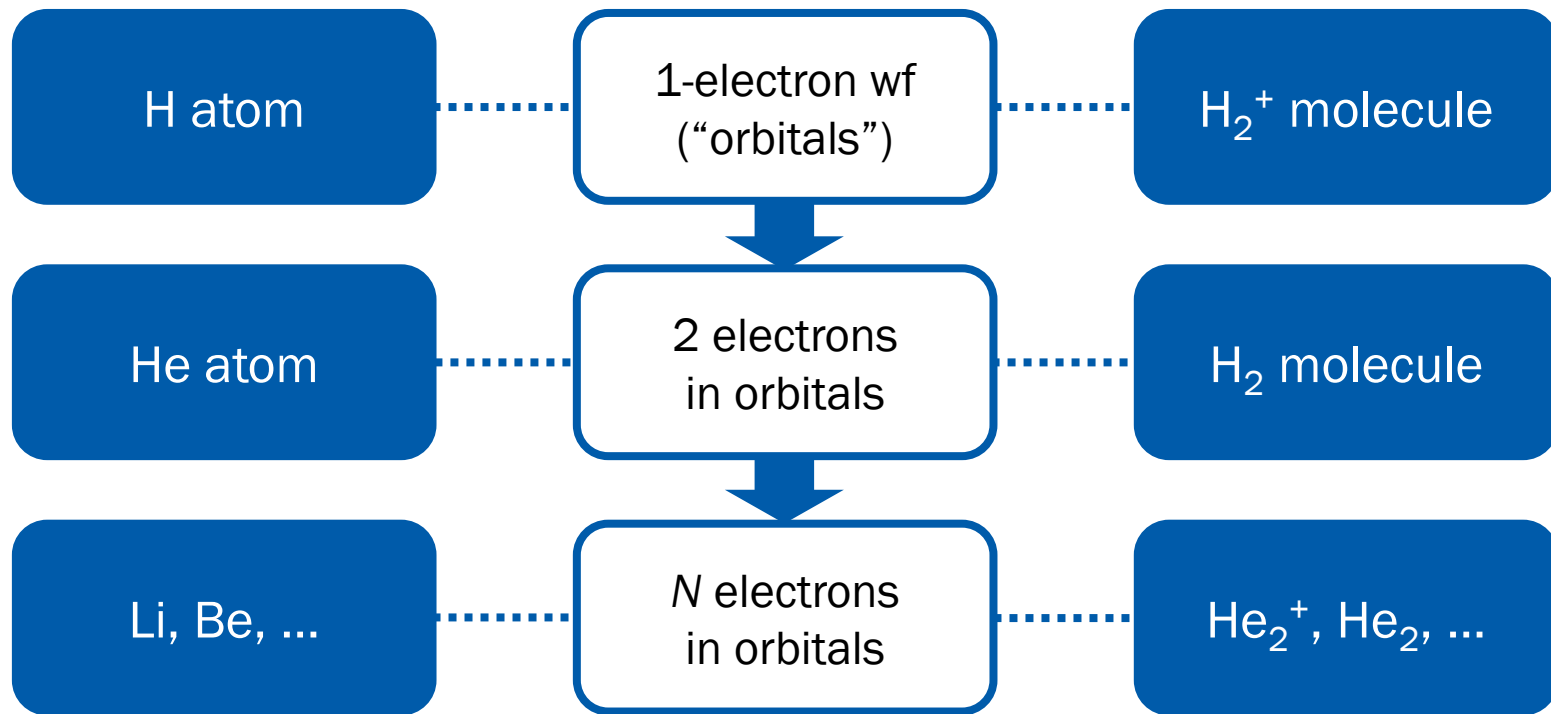
- Nuclei are **much more massive** than electrons
- Electrons adjust instantaneously to any nuclear motion
- For the fixed nuclear geometry $\vec{R}_1, \vec{R}_2, \dots$, solve the Schrödinger equation:

$$\begin{aligned}\hat{H}_{\text{el}}\psi_{\text{el}}(\vec{r}_1, \vec{r}_2, \dots; \vec{R}_1, \vec{R}_2, \dots) \\ = E_{\text{el}}(\vec{R}_1, \vec{R}_2, \dots)\psi_{\text{el}}(\vec{r}_1, \vec{r}_2, \dots; \vec{R}_1, \vec{R}_2, \dots)\end{aligned}$$

- To solve this, one can apply **any electronic structure calculation method**



Orbital Picture



Not necessarily precise (except H atom)

Last class

H_2^+ System (Variational)

- Energy

$$E_{\pm} = \frac{H_{AA} \pm H_{AB}}{1 \pm S_{AB}}$$

- Wavefunction

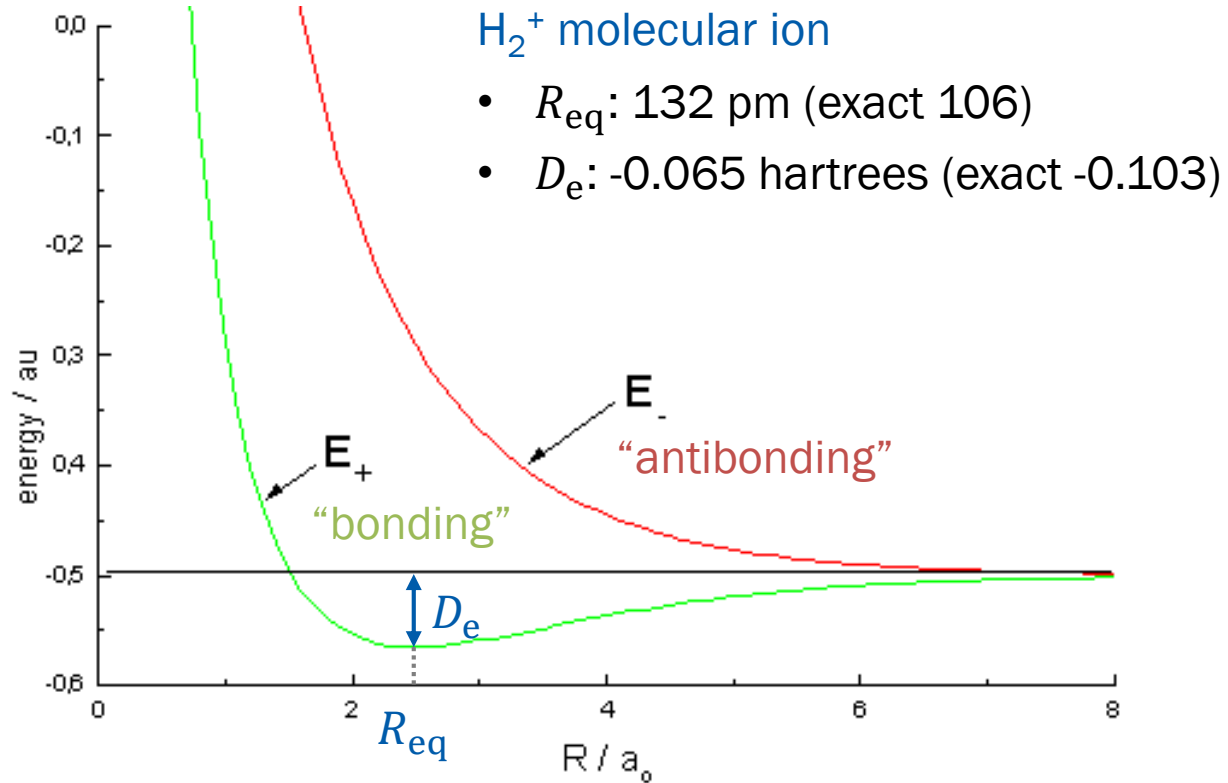
$$\phi_{\pm} = \frac{1}{\sqrt{2(1 \pm S_{AB})}} (\psi_{1s,A} \pm \psi_{1s,B})$$

where

$$H_{ij} = \int \psi_{1s,i} \hat{H} \psi_{1s,j} d\vec{r} \quad \text{and} \quad S_{AB} = \int \psi_{1s,A} \psi_{1s,B} d\vec{r}$$



“Orbital Energies”

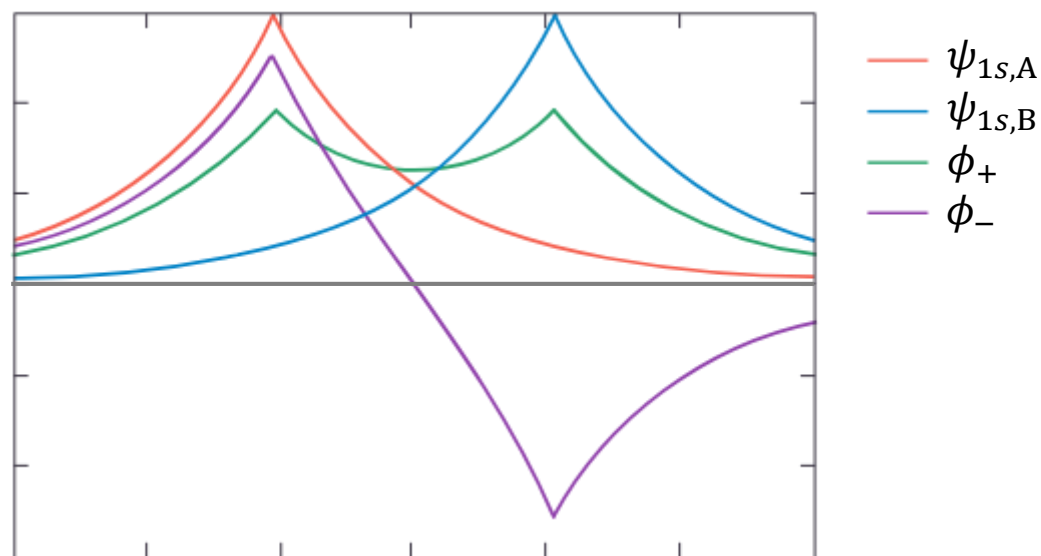


(Image: http://www.pci.tu-bs.de/aggericke/PC4e/Kap_II/H2-Ion.htm)

Last class

“Molecular Orbitals”

$$\phi_{\pm} = \frac{1}{\sqrt{2(1 \pm S_{AB})}} (\psi_{1s,A} \pm \psi_{1s,B})$$



(Image: <https://chem.libretexts.org/@go/page/1974>)

Last class

Ground-State Wavefunction for H₂

- Use H₂⁺ molecular orbitals

$$\phi_{\pm} = \frac{1}{\sqrt{2(1 \pm S_{AB})}} (\psi_{1s,A} \pm \psi_{1s,B})$$

- Slater determinant (both electrons reside in ϕ_{+})

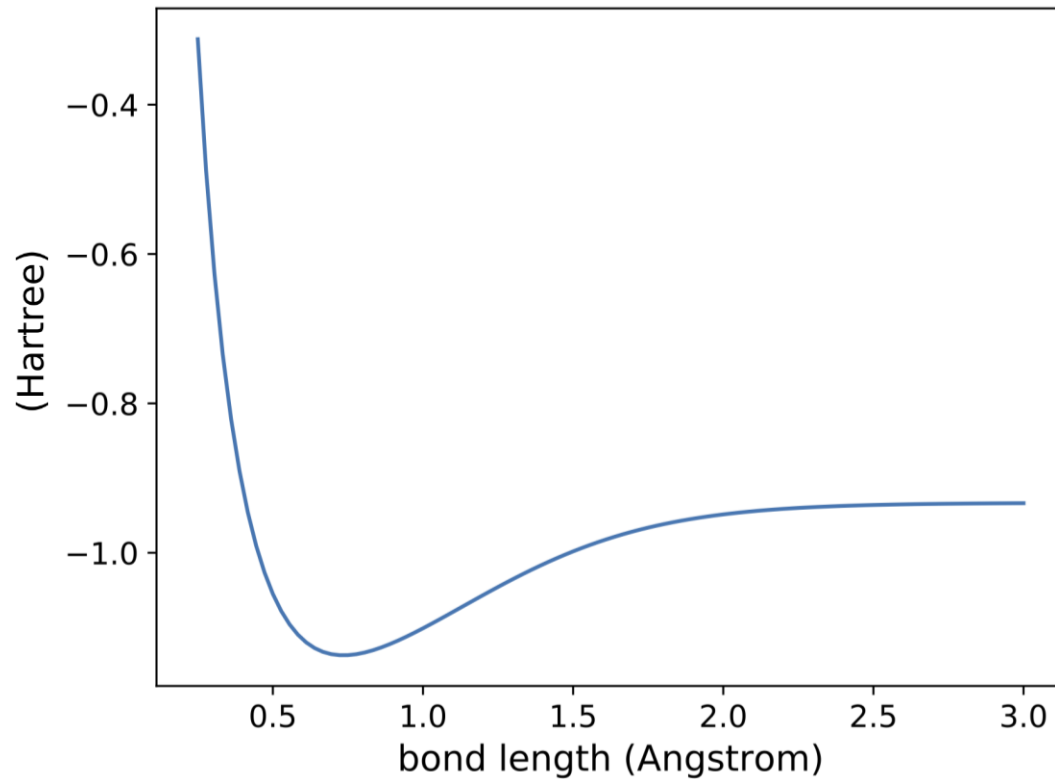
$$\psi = \frac{1}{\sqrt{2}} \begin{vmatrix} \phi_{+}(1)|\alpha\rangle_1 & \phi_{+}(1)|\beta\rangle_1 \\ \phi_{+}(2)|\alpha\rangle_2 & \phi_{+}(2)|\beta\rangle_2 \end{vmatrix}$$

$$\psi = \underbrace{\phi_{+}(1)\phi_{+}(2)}_{\text{spatial (MO)}} \times \underbrace{\frac{1}{\sqrt{2}} (|\alpha\rangle_1|\alpha\rangle_2 - |\beta\rangle_1|\beta\rangle_2)}_{\text{spin}}$$



Last class

Ground-State Energy of H_2



(Image: <https://quantaggle.com/competitions/gs-pes/>)

Last class

Valence Bond Theory

H₂ ground state = two ground-state H atoms

$$\psi_{\text{VB}} = \frac{1}{\sqrt{2(1 + S^2)}} [\psi_{1s,A}(1)\psi_{1s,B}(2) + \psi_{1s,B}(1)\psi_{1s,A}(2)]$$

We can similarly obtain **energy** by perturbation theory



MO and VB Theories

- MO theory

$$\underbrace{\psi_{1s,A}(1)\psi_{1s,A}(2)}_{\text{ionic}} + \underbrace{\psi_{1s,A}(1)\psi_{1s,B}(2) + \psi_{1s,B}(1)\psi_{1s,A}(2)}_{\text{covalent}} + \underbrace{\psi_{1s,B}(1)\psi_{1s,B}(2)}_{\text{ionic}}$$

- VB theory

$$\psi_{1s,A}(1)\psi_{1s,B}(2) + \psi_{1s,B}(1)\psi_{1s,A}(2)$$

- Better model: somewhere between MO and VB

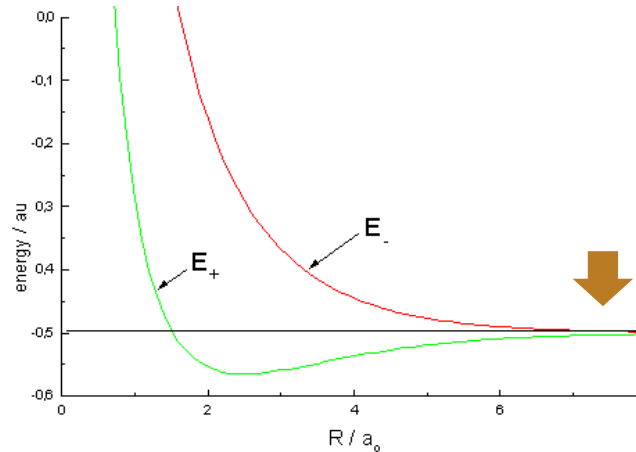
$$\psi = c_1\psi_{\text{VB}} + c_2\psi_{\text{ionic}}$$

Last class

Overlap and Chemical Bond

To make a bond from two atomic orbitals,
we need a significant **overlap** between the two

$$S_{AB} = \int \psi_A^* \psi_B d\vec{r} \neq 0$$

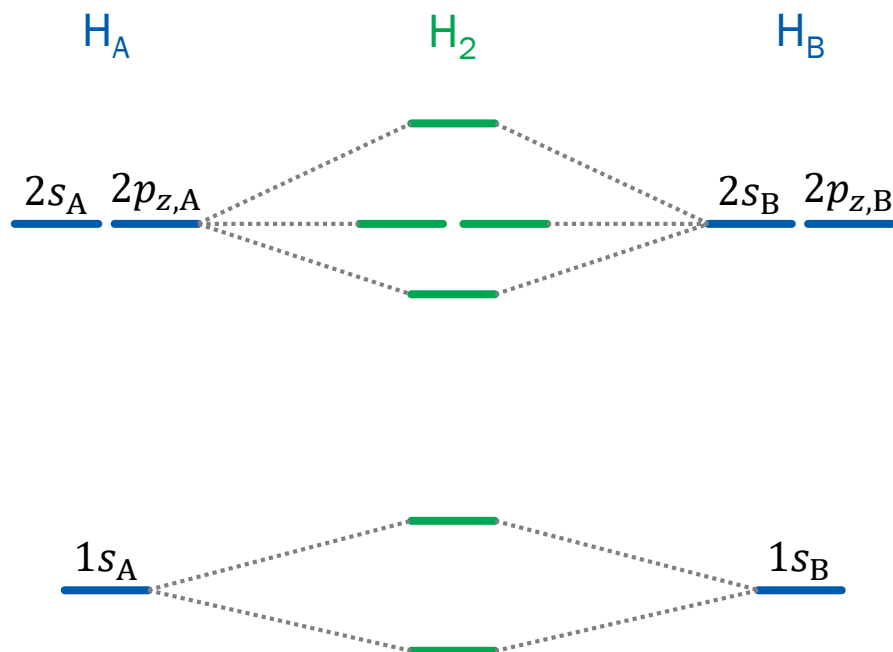


(Image: http://www.pci.tu-bs.de/aggericke/PC4e/Kap_II/H2-Ion.htm)

Last class

Molecular Orbitals

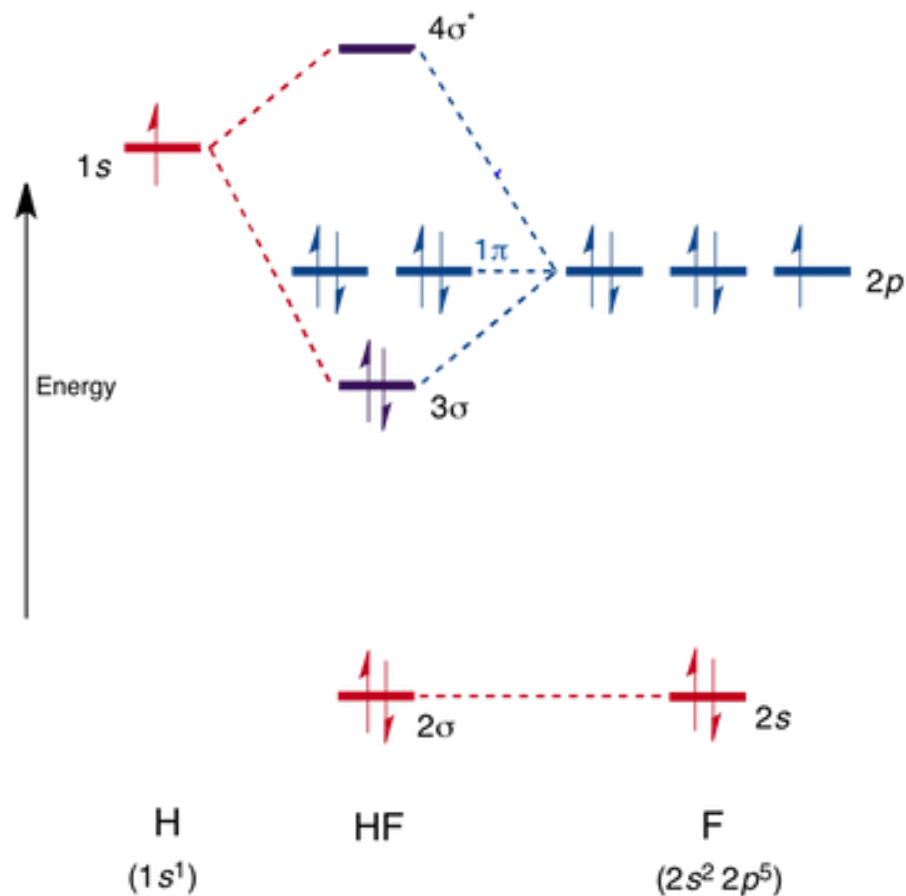
One can combine any number of atomic orbitals
to construct molecular orbitals (LCAO-MO)



* for fixed nuclear geometry

Last class

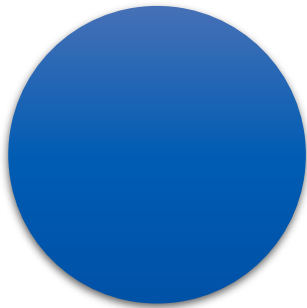
Heteronuclear Case: HF



(Image: <https://www.chemtube3d.com/orbitalshf/>)

Electronic States of Molecules

- Determines the energy levels
- Described by molecular term symbols



Atom

Spherical symmetry

$\vec{L}_{\text{total}}^2$ conserved

L is a good quantum number



Diatomic molecule

Cylindrical symmetry

$L_{\text{total},z}$ conserved

M_L is a good quantum number

Molecular Term Symbol

$$^{2S+1} [|M_L|]$$

- Symbols for $[|M_L|]$

$ M_L =$	0	1	2	3
	Σ	Π	Δ	Φ

- $M_L = m_{l1} + m_{l2} + \dots, M_S = m_{s1} + m_{s2} + \dots$
- Completely filled orbitals do not contribute

Ground State of H₂

- Electron configuration: $(1\sigma_g)^2$
- $m_{l1} = 0, m_{l2} = 0 \rightarrow M_L = m_{l1} + m_{l2} = 0$
- $m_{s1} = +1/2, m_{s2} = -1/2 \rightarrow M_S = m_{s1} + m_{s2} = 0$
- The only possible $M_S = 0: S = 0$
- Hence, the term symbol is $^1\Sigma$

Ground State of H_2^- and He_2^+

- Electron configuration: $(1\sigma_g)^2(1\sigma_u)^1$
- $m_{l1} = 0, m_{l2} = 0, m_{l3} = 0 \rightarrow M_L = 0$
- $m_{s1} = +1/2, m_{s2} = -1/2, m_{s3} = \pm 1/2$
 $\rightarrow M_S = m_{s1} + m_{s2} + m_{s3} = \pm 1/2$
- Possible $M_S = \pm 1/2: S = 1/2$
- Hence, the term symbol is $^2\Sigma$

Excited State of H_2

- Electron configuration: $(1\sigma_g)^1(1\sigma_u)^1$
- $m_{l1} = 0, m_{l2} = 0 \rightarrow M_L = m_{l1} + m_{l2} = 0$
- $m_{s1} = \pm 1/2, m_{s2} = \pm 1/2$
 $\rightarrow M_S = m_{s1} + m_{s2} = \pm 1, 0$ (doubly degenerate)
- $S = 1$ and $S = 0$ can explain possible M_S values
- Hence, the possible term symbols are $^3\Sigma$ and $^1\Sigma$

Procedure: Equivalent Electrons

- **Pauli exclusion principle:** (a, a) impossible
- **Indistinguishability:** (a, b) = (b, a)

1. Construct a table for microstates

$$M_L = m_{l1} + m_{l2} + \cdots, \quad M_S = m_{s1} + m_{s2} + \cdots$$

2. Eliminate the states that violate the two conditions

$$\text{Total number of microstates} = G!/[N!(G - N)!]$$

3. Deduce the possible values of S and $|M_L|$ (start from largest M_S and then largest M_L)



B₂ Molecule

$$(1\sigma_g)^2(1\sigma_u)^2(2\sigma_g)^2(2\sigma_u)^2(1\pi_{ux})^1(1\pi_{uy})^1$$

Possible $M_L = 0, \pm 2$, possible $M_S = 0, \pm 1$

		M_S		
		1	0	-1
M_L	2		$(1^+, 1^-)$	
	0	$(1^+, -1^+)$	$(1^+, -1^-); (-1^+, 1^-)$	$(1^-, -1^-)$
	-2		$(-1^+, -1^-)$	

Total number of microstates = $4!/(2!2!) = 6$

Deducing S and $|M_L|$

		M_S		
		1	0	-1
M_L	2			
	0		 	
	-2			

- $M_S = 0, \pm 1$ and $M_L = 0 \rightarrow S = 1$ and $|M_L| = 0$ ($^3\Sigma$)
- $M_S = 0$ and $M_L = \pm 2 \rightarrow S = 0$ and $|M_L| = 2$ ($^1\Delta$)
- $M_S = 0$ and $M_L = 0 \rightarrow S = 0$ and $|M_L| = 0$ ($^1\Sigma$)

Ground State of B₂

For the electron configuration

$$(1\sigma_g)^2(1\sigma_u)^2(2\sigma_g)^2(2\sigma_u)^2(1\pi_{ux})^1(1\pi_{uy})^1,$$

we have three possible term symbols: ${}^3\Sigma$, ${}^1\Delta$, and ${}^1\Sigma$

According to Hund's rule 1, the ground state is ${}^3\Sigma$.

Symmetry of Term Symbol

- g/u : sign change upon inversion

- can be obtained from molecular orbitals

$$g \cdot g = g, \quad g \cdot u = u, \quad u \cdot u = g$$

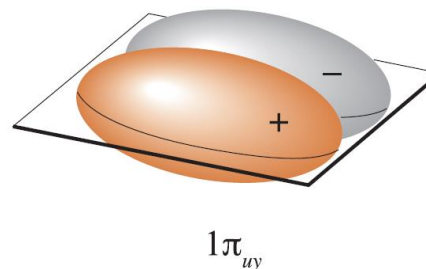
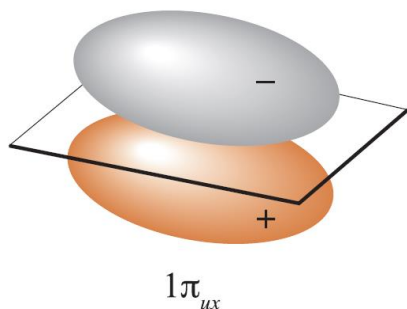
- $+/-$: sign change upon reflection through an internuclear plane (only applies to Σ electronic states!)

- can be obtained from molecular orbitals

$$(+)(+) = (+), \quad (+)(-) = (-), \quad (-)(-) = (+)$$

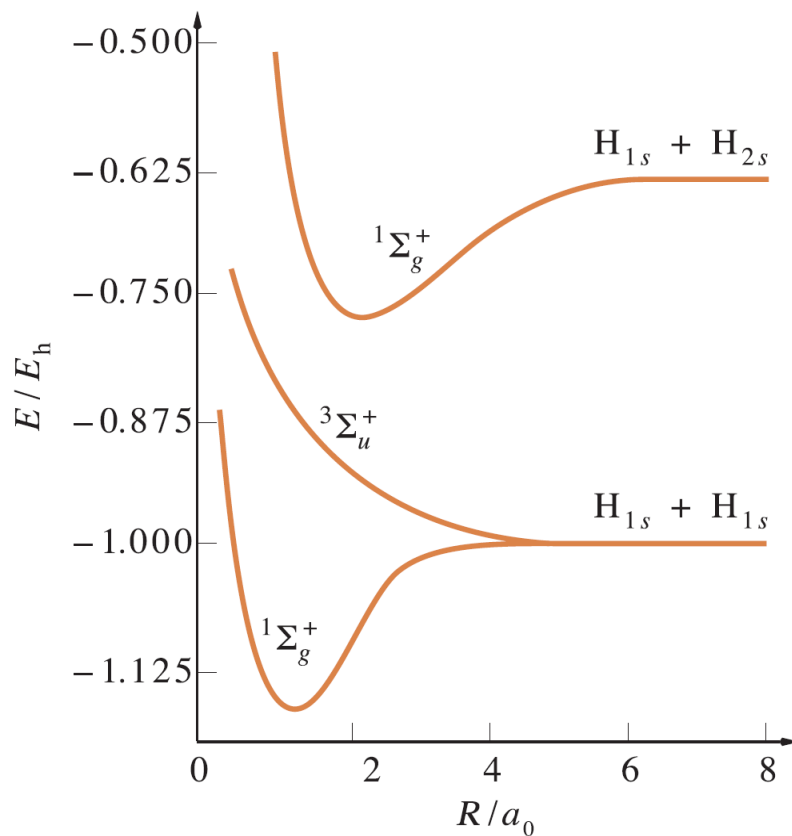
Example of B₂

$$(1\sigma_g)^2(1\sigma_u)^2(2\sigma_g)^2(2\sigma_u)^2(1\pi_{ux})^1(1\pi_{uy})^1$$



- Both orbitals are *u*, so $u \cdot u = g$
- One is (+) and the other is (-), so $(+)(-) = (-)$
- Hence, the ground-state term symbol is ${}^3\Sigma_g^-$

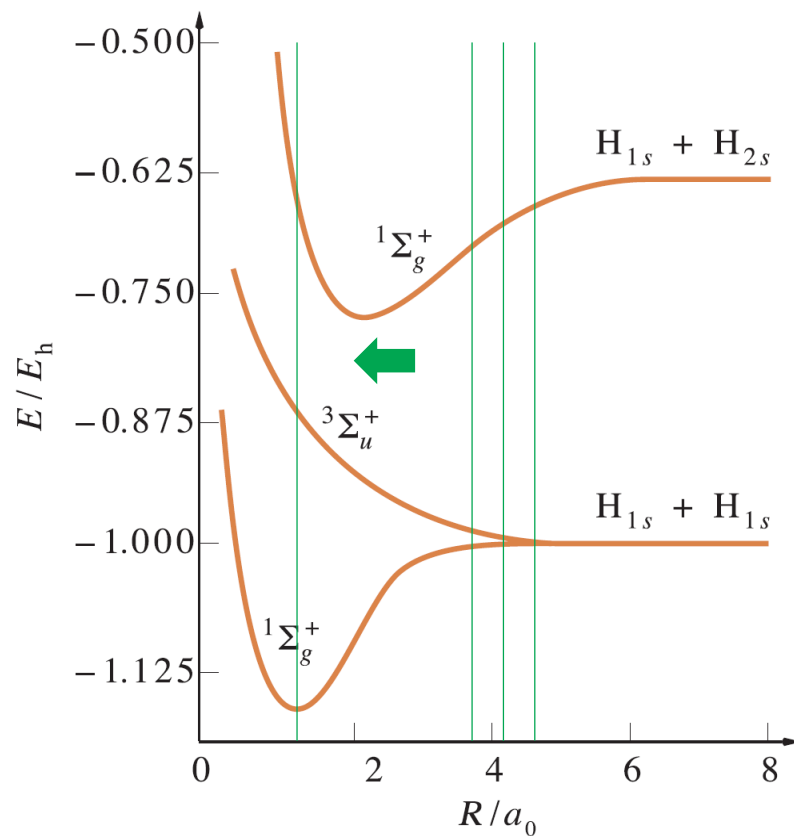
Potential Energy Curves



(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Figure 11.12)

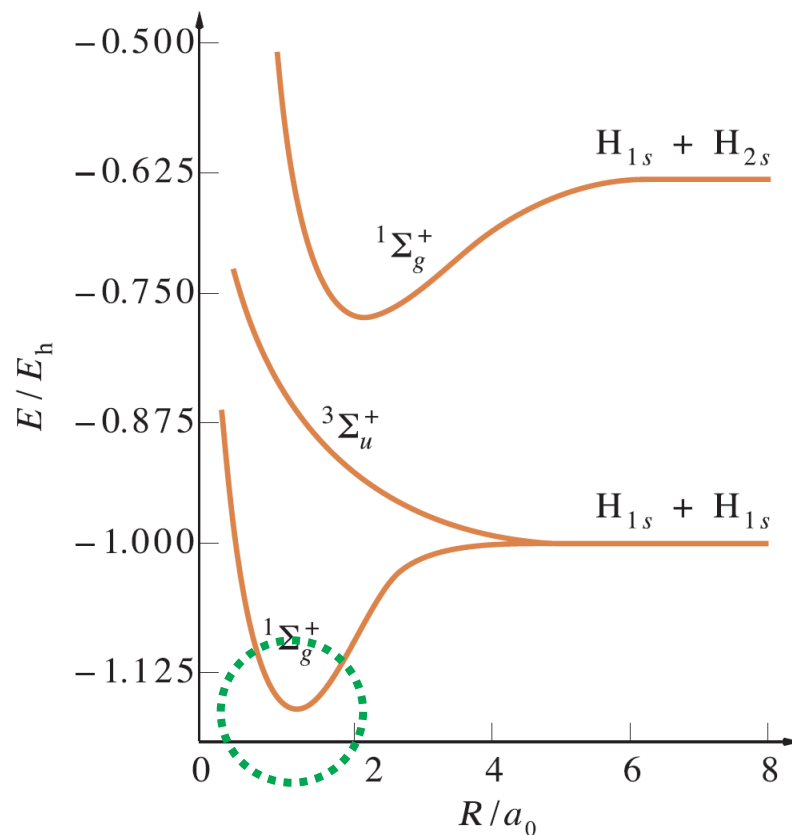
Not in text

Geometry Optimization



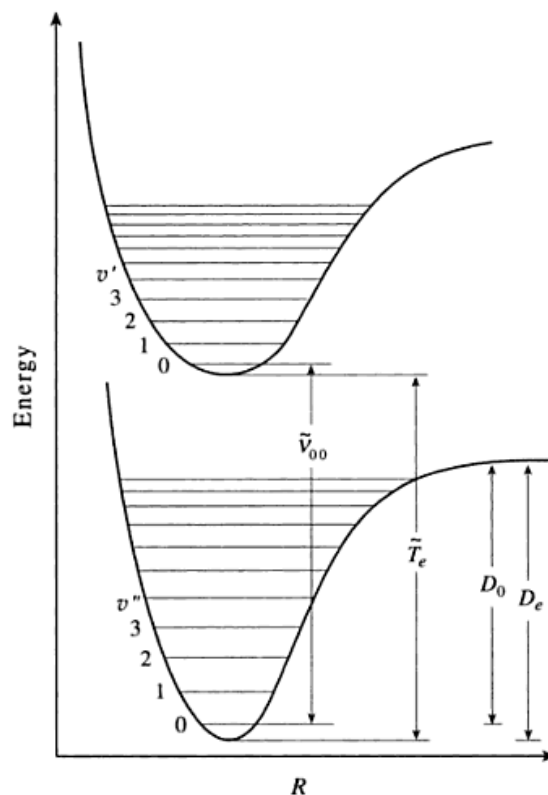
(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Figure 11.12)

Origin of Molecular Vibrations



(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Figure 11.12)

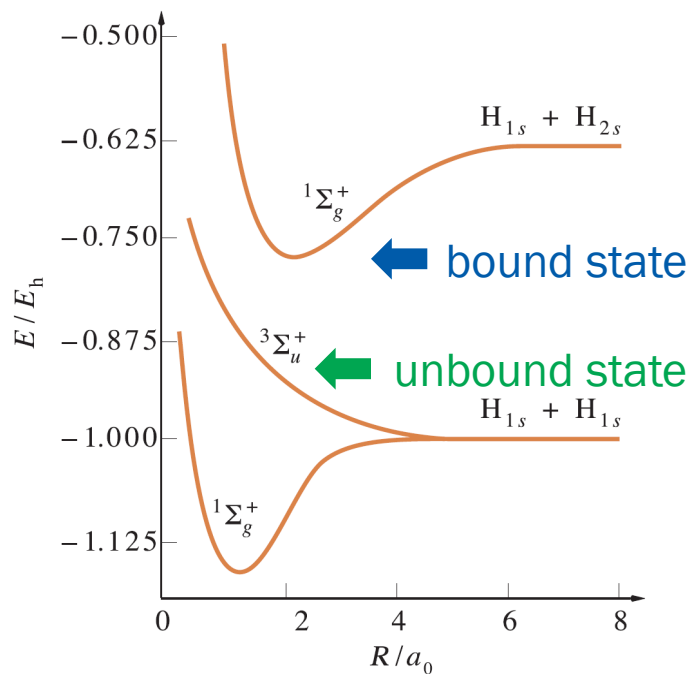
Vibrational Levels



Not in text

Fate of Excited System

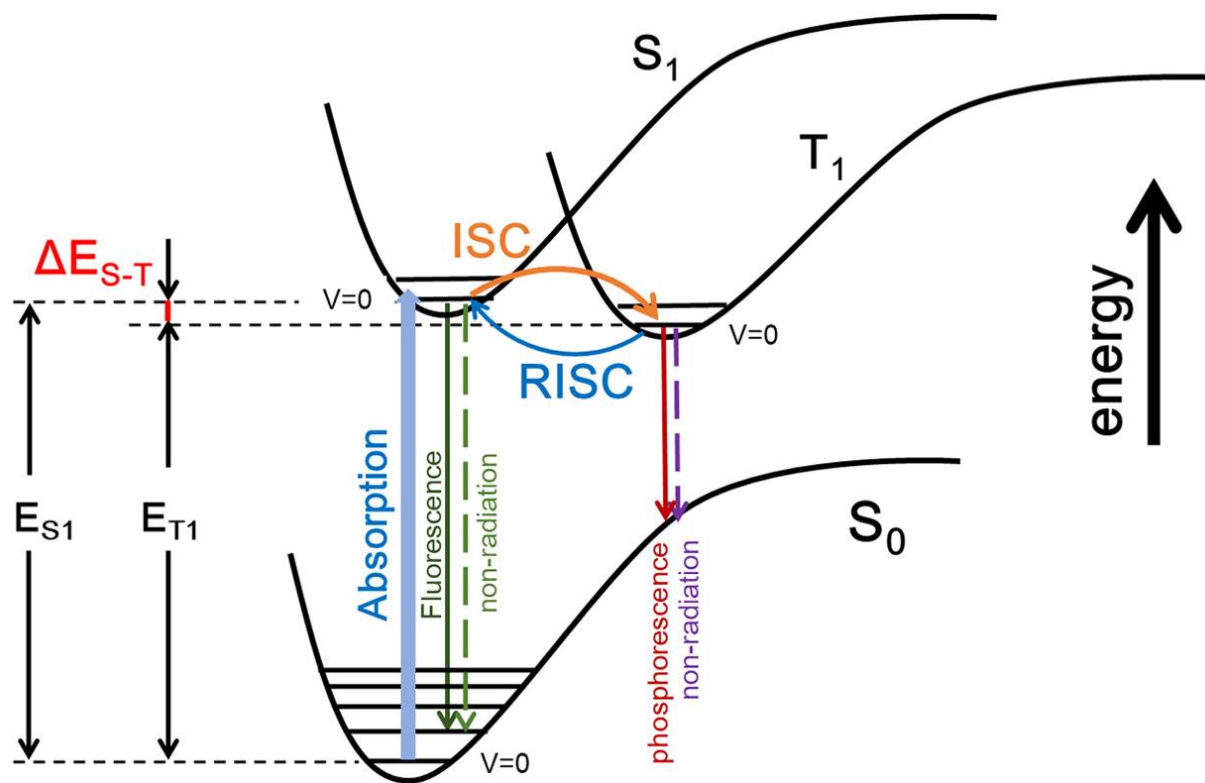
depends on the excited state: **bound** or **unbound**?



(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Figure 11.12)

Not in text

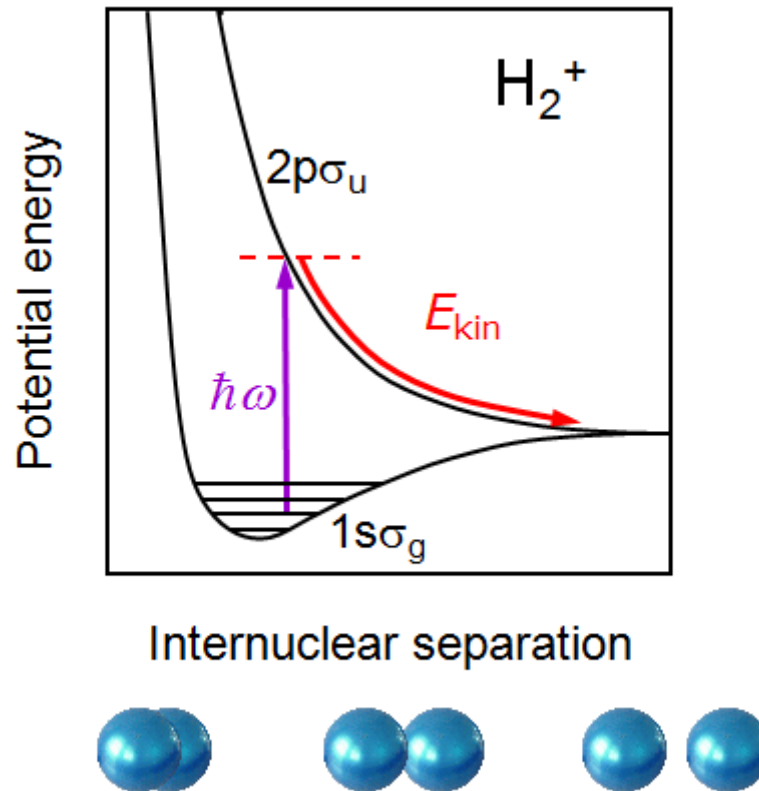
Fluorescence and Phosphorescence



(Image: Zheng et al., *J. Phys. Chem. Lett.*, 2022)

Not in text

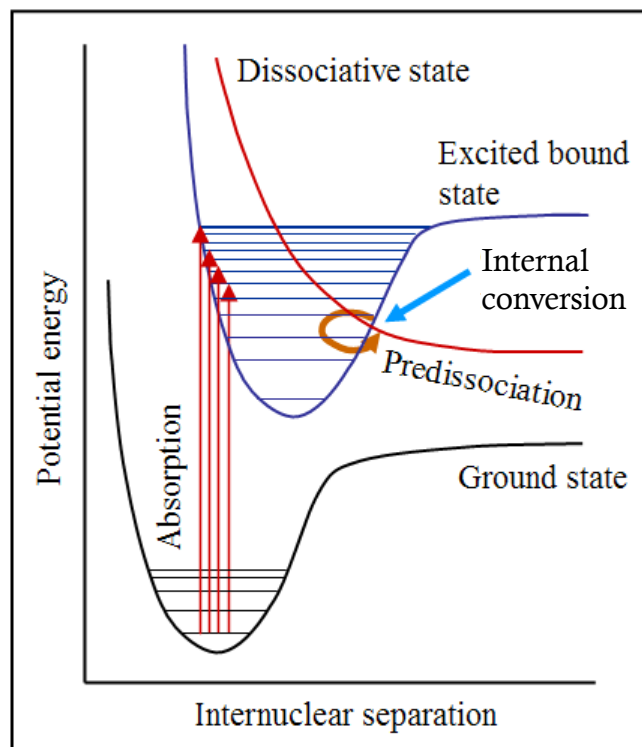
Photodissociation



(Image: https://commons.wikimedia.org/wiki/File:Hydrogen_molecular_ion_dressed_in_photons.png)

Not in text

Predissociation



(Image: Iqbal, Ph.D. Dissertation, 2010)